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TACTILE EDGES AND MOTION VIA PATTERNED MICROSTIMULATION OF THE HUMAN SOMATOSENSORY CORTEX

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Abstract

Intracortical microstimulation (ICMS) of somatosensory cortex evokes tactile sensations whose properties can be systematically manipulated by varying stimulation parameters. However, ICMS currently provides an imperfect sense of touch, limiting manual dexterity and tactile experience. Leveraging our understanding of how tactile features are encoded in the primary somatosensory cortex (S1), we sought to inform individuals with paralysis about local geometry

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and apparent motion of objects on their skin. We simultaneously delivered ICMS through electrodes with spatially-patterned projected fields (PFs) evoking sensations of edges. We then created complex PFs encoding arbitrary tactile shapes and skin indentation patterns. By delivering spatiotemporally-patterned ICMS, we evoked sensation of motion across the skin, the speed and direction of which could be controlled. Thus, we improved individuals' tactile experience and use of bionic hands.

Introduction

Touch conveys rich information about our interactions with objects that facilitates dexterous behaviors(1). When we touch and manipulate an object, we identify its textural qualities and local geometric features. This information is consolidated into a three-dimensional representation of the object in a process known as stereognosis(2). This is enabled, in part, by neurons in primary somatosensory cortex (S1) that have receptive fields tuned for these features(3). Spinal cord injury often disrupts both motor and sensory functions of the hand(4). This loss can be partially circumvented by restoring hand and arm movements using robots controlled by motor intent signals decoded from motor cortex(5); however, normal dexterous hand function is impossible without somatosensation. Intracortical microstimulation (ICMS) of Brodmann's Area 1 (BA1), a subregion of S1, has been shown to evoke vivid tactile sensation seeming to originate from the skin (projected fields, PFs)(6–8) and provides a promising avenue for restoring somatosensory feedback(9–11).

Initial ICMS attempts have focused almost entirely on two features: sensation location(6–8, 12, 13) and intensity(10, 14). ICMS can be used to intuitively signal the location of object contact on a bionic hand by linking sensors on the hand to electrodes in BA1 which evoke somatotopically-matched sensations(9). The perceived intensity, controlled by ICMS amplitude(6) or frequency(11) can be used to signal interaction force. However, the sense of touch is far richer than these two components. It also conveys information about textures, material properties, local contours, and motion of objects across the skin. These aspects of touch are critical for our body perception, and ability to dexterously interact with objects.

Here, we investigated whether ICMS could encode local geometrical features of touch (Fig.S1A). We leveraged the fact that overlapping PFs created with multi-electrode ICMS (mICMS) combine additively(12) to create composite PFs with specific shapes. We temporally patterned mICMS to evoke a sense of motion across the skin. Together, this spatiotemporal patterning expanded the repertoire of artificial touch beyond the basic features.

Spatial patterning of ICMS evokes tactile shape percepts.

We performed a series of experiments in two participants (c1-c2) with cervical spinal cord injuries, using microelectrode arrays implanted in the hand representation of BA1 (Fig.1A,B). In these experiments, we identified combinations of three electrodes with spatially-aligned PFs which we hypothesized would evoke edge-like sensations. Indeed, the participants, without knowledge of the intended goal of the experiment, spontaneously reported feeling edges with specific orientation and shape (MovieS1). To test the robustness

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of this phenomenon, we repeated this paradigm with various PF configurations (Fig.2A) and found that both participants could reliably discriminate the edge orientation regardless of the tested digit (Fig.2B, c1:81-89-84% on d1-d2-d3, n=150; c2:61–68% on d2-d3, n=75; $p<0.05$ for all).

Next, we assessed the effect of ICMS duration and amplitude on orientation recognition. Consistent with natural touch(15, 16), increased duration improved orientation recognition, reaching asymptotic performance at 0.5 seconds (78%, n=225, $p<0.05$) (Fig.S2A). This implies that this restored feature is sufficiently rapid to enable real-time feedback in bionic devices(17). Unlike duration, there was no effect of amplitude on discriminability (Fig.S2B, n=270, $p>0.05$), also consistent with natural touch(18). Thus, the spatial properties of the artificial touch are invariant to intensity.

We extended this paradigm to arbitrary orientations and simple shapes, asking participants to identify the shapes evoked by two-PFs (3 or 5 shapes for c2 and c1, respectively). Both participants were able to correctly identify shapes on multiple digits (Fig.2C,D, c1:82±9–72±8%, n=200, c2:50±14–48±8%, n=120, for d2-d3, $p<0.05$), though they were tended to confuse similar shapes (Fig.S3A). Next, we tested shapes composed of three PFs (Fig.S3B). c1 completed the task although slightly worse due to its greater complexity (60±22.7–57±7.6% for d2-d3, $p<0.05$, n=160, Fig.2E, Fig.S3C), as observed in natural touch(16).

Finally, we examined the participants' drawings of these perceived shapes (Fig.S3A, Fig.S4) and found that both the length and orientation of the evoked tactile shape matched our predictions from the single electrode reports (Fig.S3D–F). Participants would often report sensations of a continuous edge even when the individual electrode PFs were discontinuous (Fig.S3A,B, Fig.S4) an observation that highlights the similarity between the ICMS 2-point discrimination threshold and that of natural touch(19). This is also in line with the tactile funneling illusion(20, 21), where stimulation of multiple electrodes (like mechanical stimuli on the skin) can be spatially-integrated and cause a single cortical activation zone.

The results presented so far involve simple flat/planar sensations; however, most objects are rich with tactile features detected by cutaneous mechanoreceptors(22). We modulated the amplitude and timing of the stimulation of electrodes with adjacent PFs (Fig.3A) resulting in c1 reporting sensations of rounded edges (MovieS2). To test the precision with which c1 could perceive this curvature, we modulated the amplitude to simulate changes in curvature. c1 could reliably identify (>75% success) curved stimuli once this amplitude difference exceeded 20% (Fig.3B), marginally worse than natural touch(23, 24), though performance depends on the contact area between the surface and finger. Above this threshold, c1 could successfully identify randomly-presented concave, flat, and convex surfaces (Fig.S5, 64±7%, n=75, $p<0.05$).

When we grasp an object, cutaneous inputs from multiple digits are integrated to form an object representation(25). To determine if our approach could be extended to support this complexity, we used mICMS that included PFs on multiple digits. Participant c1 reported sensations recalled from grasping real objects (MovieS3). Depending on their relative locations, the combination of three edges encoded on 9 electrodes with PFs on multiple

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digits evoked various natural-contact sensations: a pen when the three encoded edges on the three fingers were colinear across the digits, a can (grasped from the top) when the edges were colinear all along the length of the digits, and a ball, when the PFs evoked on multiple digits all overlapped (no alignment) (Fig.3C). To test whether these evoked sensations were reliable, we designed a multichoice task in which we randomly encoded 3D-objects via mICMS. Participant c1 successfully identified the encoded objects above chance (79±11%, n=45, $p<0.05$). Our findings demonstrate that informed mICMS patterns could be used to evoke vivid sensory experience related to real-life objects.

Spatiotemporal patterning of ICMS evokes sensations of tactile motion.

Static spatial information, as described above, could be used to improve object recognition for people with impaired somatosensation. However, stereognosis requires active manipulation and contact motion across the fingertips to build a rich object representation. In both vision and touch, information about motion is extracted from the spatiotemporal activation pattern across a two-dimensional sensory sheet (retina(26) and skin(3)). In natural touch, the perceived motion direction is determined by the ordered responses of a subpopulation of cortical neurons in BA1(3). We tested if we could mimic this effect by stimulating electrodes with adjacent PFs in a specific temporal sequence (continuous cortical trajectory) forming along proximal-distal (along the finger) and radio-ulnar (across the finger) axes (Fig.4A, Fig.S7A). Upon stimulation, both participants immediately described the evoked sensation as motion on their skin (MovieS4), and were able to distinguish the four motion directions (c1:76±14%, n=180, c2:78±15%, n=60; $p<0.05$, Fig.4A, Fig.S7A top), even when stimuli spanned multiple digits (c1:98±2%, n=120; c2:75±14%, n=75, $p<0.05$, Fig.4A bottom, Fig.S7B). We were able to evoke other apparent motions in participant c1, that included circular (96±7%, n=90, $p<0.05$) and radial motion (81±17%, n=75, $p<0.05$, Fig.S7D,E).

Next, we assessed the effect of ICMS duration and amplitude on the participants' ability to determine direction of motion. Whereas duration had a significant effect (50ms–30%, 200ms –72%, n=60, $p<0.05$, Fig.S6A, with fixed inter-train-spacing), similar to natural touch(27, 28), amplitude did not ($p>0.05$, n=60, Fig.S6B). Similar to the results on orientation described above (Fig.S2A), only a few hundred milliseconds were required to interpret motion direction, making this practical for closed-loop brain-computer interfaces (BCIs). Apart from their spatial arrangement on the skin, both edge orientation and motion encoding were achieved regardless of the specific characteristics of the individual PFs (quality, geometry, size, or perceived intensity).

To understand how multiple ICMS trains might influence motion percepts, we systematically varied their relative timing (negative or positive inter-train-intervals) and asked the participants to report whether a perception was continuous or pulsatile (Fig.4B). Perceived motion occurred only when trains were separated by 0 to 0.5 seconds. Any overlap caused them to appear to be a single event at two locations, whereas more than 0.5 seconds between trains, typically caused the mICMS to be perceived as two successive events (intermittent motion(29)). This result occurred regardless of the PF separation, and the timings closely match those of natural touch(29, 30).

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When we manipulate an object, neurons with the receptive field on the contact area start to fire and the object's motion is encoded via temporal modulation of the firing(31, 32). To mimic this process, we investigated more effective dynamics of ICMS like those proposed for intracortical visual prostheses(33) and cochlear implants(34). In c1 we simultaneously stimulated three electrodes with aligned PFs and then modulated the voltage fields (via ICMS-amplitude) across these three electrodes (VFS, Fig.4C) moving the peak in the direction of the desired apparent motion. The direction of change of the current ratio across electrodes determines in which direction the pattern is "moving" on the cortex and how dynamic the pattern is perceived to be. With this spatiotemporal ICMS modulation, c1 was able to identify motion among four directions on the skin of d2 (c1: 74±12%, $n=160$, $p<0.05$, MovieS5) similar to sequential mICMS (NO-VFS, $p>0.1$). This encoding strategy might also evoke patterns of neural activation that are more similar to those generated by natural tactile motion. Information about direction and speed of motion is transmitted by local transients of neural firing modulated across neurons with adjacent receptive field(3), replicated here with modulation of mICMS amplitude. In the field of somatosensory neuroprosthetics, the use of more biomimetic approaches to stimulate the nervous system showed benefits compared to canonical strategies(10, 35, 36). This result demonstrates that this encoding method enables the simultaneous disentanglement of three distinct tactile features from ICMS: location, intensity, and motion direction.

The speed variation, that occurs during slip or surface exploration, is fundamental for tactile motion encoding(1). Hence, we varied the temporal activation of two adjacent PFs, and asked the participants to identify which of two mICMS patterns they perceived to be faster (Fig.4D, MovieS6). Both participants were able to discriminate motion speed with Weber Fraction=0.56±0.09, which falls in the same range as natural touch(37). Changing the onset lag between two equal-length ICMS trains caused the perceived speed to vary, when presented on both d2 and d3 in participant c1 (Weber Fraction=0.63±0.06, Fig.S8B). To ensure that the participants were identifying motion speed rather than train duration, we repeated the task using electrode pairs with different distances between PFs (Fig.S8A). We interleaved channel pairs with short (~0.6cm) and long (~1.4cm) spatial separation on d2 in participant c1. There was no significant difference in point-of-subjective equality (PSE) for either the standard (500ms) or comparison (80–920ms) stimuli when presented for a given electrode pair. In contrast, ΔPSE was 360ms (PSE_speed=1.63cm/s) and −140ms (PSE_speed=1.67cm/s) for short and long PF separation, respectively, confirming that the effect is one of perceived speed, not train duration (Δdistance/ICMS_duration, cm/s). This speed ICMS encoding scheme resembles the afferent spatiotemporal code of motion on the skin surface(38, 39).

Tracing trajectories on the cortex allows complex tactile shapes to be encoded.

Motion is also an important component of tactile discrimination. Complex tactile patterns are easier to recognize when they are drawn on the skin (called graphesthesia) rather than simply indented(40) and active exploration of objects is more informative than static deformation of the skin(41). Similarly, shapes traced through the visual field using sequential mICMS within V1 are better recognized than using simultaneous mICMS(33). To test the effect of motion on complex tactile discrimination, we activated PFs representing

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letters either sequentially or simultaneously. Participant c1 was able to identify the letters T,L,C,O and I represented on 3–6 adjacent PFs (Fig.5A) with both simultaneous and sequential mICMS, though performance was significantly higher in the sequential condition (49±17% versus 37±21%, n=150, $p<0.05$, Fig.5B). The advantage of sequential activation increased with shape complexity (Fig.5C). In principle, the outlines of other shapes could be traced using the same approach.

Sequential activation also has the benefit of reducing the current injected into the cortical tissue at any given point, which might decrease sensory adaptation(33). With simultaneous stimulation, when the complexity of the tactile shape increases the resulting PFs can interact, often coalescing into a single PF that is not always described as a specific shape (Fig.5C). Sequential mICMS seems to diminish the likelihood of spatial blurring, improving the effective spatial resolution of the evoked patterns.

Finally, we tested whether these ICMS-based sensations could be exploited in the brain-control of bionic arms in a real-time, functional task. We asked participant c1 to use a bionic arm via brain control to steer a wheel in response to movement perturbations (left, right, none) encoded via mICMS (Fig.5D, MovieS7). The participant was able to actively counteract the perturbations, exploiting only the ICMS-based sensations in real-time (No-CL:85±9%, n=40, $p<0.01$, Fig.5E). Moreover, participant c1 accomplished the task even with additional cognitive loads given by the simultaneous execution of mental tasks (Low-CL:80±12%, High-CL:50±21%, n=40, $p<0.01$, Fig.5E, MovieS7).

Discussion

This study advances our understanding of the effects of neurostimulation toward the goal of building better cortical neuroprostheses able to restore close-to-natural sense of touch. Despite the fact that BA1 neurons are often tuned to specific features (like orientation/motion direction), ICMS has been typically used to evoke only simple tactile percepts, similar to the phosphenes evoked with cortical visual prostheses(42). This limitation is likely the result of activating all neurons in the electrode proximity regardless of their specific stimulus selectivity(43), resulting in a non-specific pattern. However, despite this, mICMS can be used to evoke coherent visual forms(44–46), and now coherent tactile percepts.

Our participants often perceived edges even when the individual PFs had spatial gaps. Illusion studies have revealed that top-down sensory expectations can modify the sensory experience(47), perhaps in this case to create a sense of continuity between individual PFs or the percept of motion from discontinuous PFs.

Our findings show that the perceived edge length correlated with the separation (cortical distance) between electrodes (Fig.S9A). This means that edge length and orientation are functions not only of the number of stimulated electrodes, but also of their spatial distribution. Furthermore, the direction of motion results from the cortical trajectories of mICMS-induced activity (Fig.S10A–C).

The sense of touch also allows three-dimensional shapes to be perceived(25). Tactile object perception includes the integration of information including contact points, curvatures,

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texture and local edges that allows objects to be identified, even in the absence of vision(24). Previous studies have attempted to restore a sense of object shape, compliance or size using electrical stimulation of nerves(48, 49) and cortex(10, 50). Here, we advance our knowledge by exploring three dimensional shape encoding that leverages tactile features evoked on multiple digits. We hypothesize that when coupled with visual feedback and with active manipulation, these artificial sensations will result in even more natural and accurate percepts due to the additional context and the spatial processing capacity of the tactile system(15).

mICMS has previously been presented as a promising approach for improving other properties of artificial touch. Reductions in detection thresholds and reaction times have been reported(13, 51), as well as improved tactile discriminability and localizability(10, 12). However, a potential drawback of mICMS is its effect on motor decoding; S1 ICMS directly trigger activity in the primary motor cortex (M1) (52) disrupting decoders. This disruption, however, can be minimized by implementing biomimetic neurostimulation, stimulus waveforms which emphasize contact transients over maintained contact(9, 53).

Stimulation hardware can also be substantially improved. Our S1 arrays consisted of a 6×10 matrix electrodes, all of the same length, covering 2.4×4.0mm, thereby allowing only limited coverage of the cortical representation of the hand. Consequently, the combinations of perceptual edge lengths, orientations, and locations which can be evoked are severely limited. Closer electrode spacing would allow for more detailed maps of PFs that are necessary to restore acuity and richness closer to natural touch.

The fact that the digit representation is more accessible in BA1 than in other regions (like 3b and BA2) with the approved microelectrode technology (Fig.S1B), made it an ideal candidate for BCI applications. Indeed, though we might achieve better spatial resolution by targeting the smaller Area 3b, its position deep in the central sulcus poses surgical and technological challenges. Furthermore, BA1, has larger, more complex receptive fields, typically encoding stimulus features that include global motion (vector average mechanism(3)) and shape. Moreover, in BA1, texture information carried in patterns of peripheral afferent activation are converted into a rate code by the spatial and temporal filtering properties of cortical neurons(54). Implementing specific spatiotemporal mICMS patterns could eventually allow us to encode complex sensations related to textures.

Finally, simultaneous recording of the evoked neural activity during these ICMS patterns, combined with the reported sensory experiences, could be a powerful tool to improve the prediction of the expected mICMS outcomes.

Although we successfully implemented a brain-controlled bionic arm with movement sensations, to expand the repertoire of artificial touch convey local edges and motion with mICMS, bionic hands require better sensorization, including smaller sensors and the ability to detect motion or tangential forces (using highly-sensorized artificial skin(55, 56)).

Despite the multidimensionality of tactile experience, we show that by using principled neuroscientific understanding of the neural encoding of conscious sensory experiences, it is possible to produce complex tactile sensations even within the limitations of existing neural

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interface technology. With further development of higher-resolution electrodes and sensors, our stimulation protocols have the potential to restore natural touch to people with paralysis.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Competing interests:

N.G.H. and R.A.G. serve as consultants for Blackrock Neurotech, Inc. R.A.G. is also on the scientific advisory board of Neurowired. LLC., M.L.B., J.L.C., and R.A.G have received research funding from Blackrock Neurotech, Inc. though that funding did not support the work presented here. P.W. served as a consultant for Medtronic. A.R.S. served as a consultant for Google DeepMind. S.J.B., C.M.G., G.V., R.A.G., and R.L., are inventors on a pending international (PCT) patent application submitted by The University of Chicago and The University of Pittsburgh that covers multi-channel microstimulation of the somatosensory cortex (n. 23-0518-WO).

Data and materials availability:

All data stored at the Data Archive BRAIN Initiative: https://dabi.loni.usc.edu/dsi/S8Q532OTAXMS. The code is available here: https://github.com/CorticalBionics/EdgesAndMotion.git

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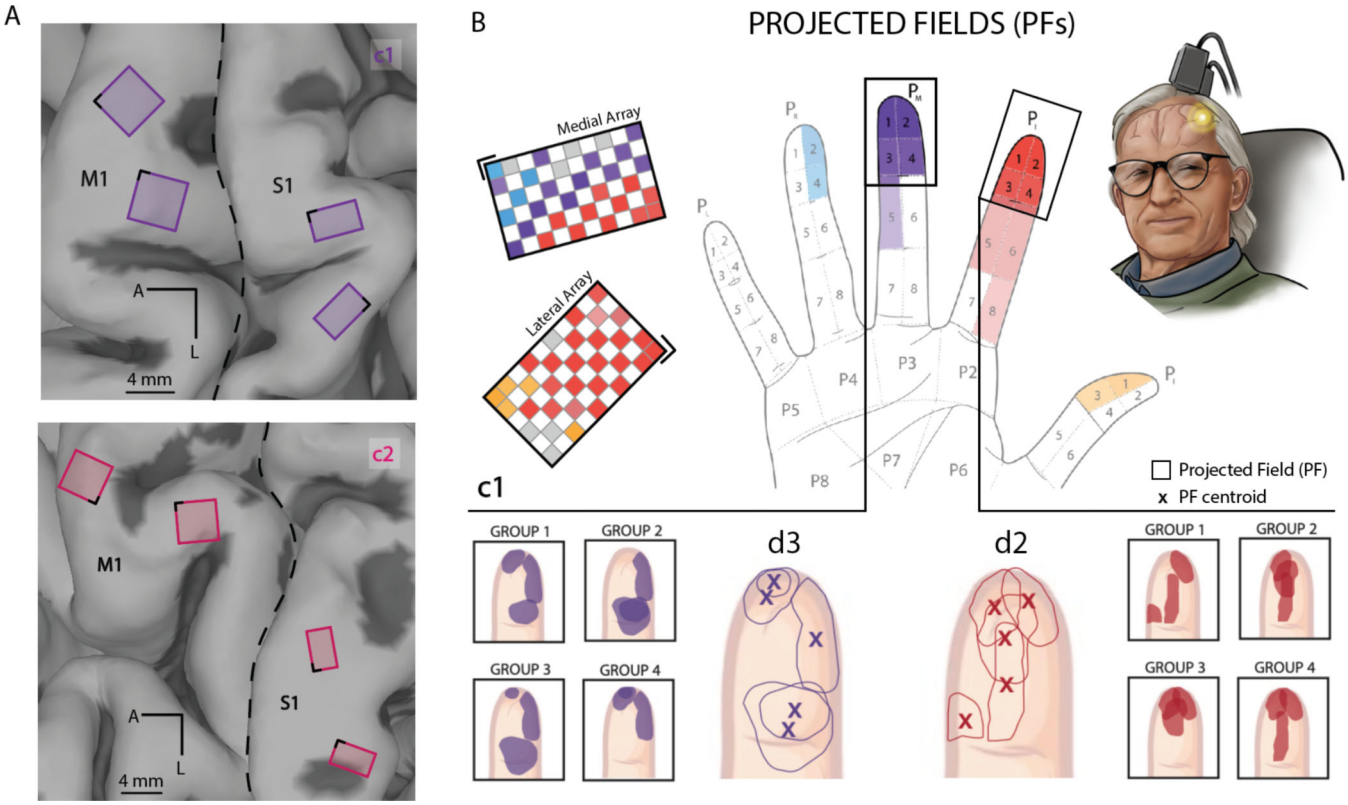


Figure 1. ICMS of S1 evokes tactile projected fields whose location follows the expected somatotopic organization.

A| Four Utah arrays (Blackrock Neurotech, Inc.) were implanted in participants c1 and c2, two of which were placed in the hand representation of Brodmann's area 1 (a subregion of S1), based on localization with brain imaging techniques (fMRI and/or MEG). Here we show array locations based on intra-operative photos, superimposed on a pre-surgical anatomical MRI. The central sulcus is indicated by the dashed line. M1: motor cortex. S1: somatosensory cortex. L: Lateral. A: Anterior. **B |** Locations of projected fields. The location on the hand where sensation is experienced, for each S1 channel for participant c1. The top and bottom arrays are medial and lateral, respectively. Colors denote the locations of the projected fields. Gray squares denote electrodes that evoke sensations on the dorsum of the hand, and white squares denote unwired electrodes. Black corners indicate alignment. Zoomed images of fingertips show precise locations of single electrode PFs on d2 and d3 and the possible combinations (groups) tracing different tactile shapes.

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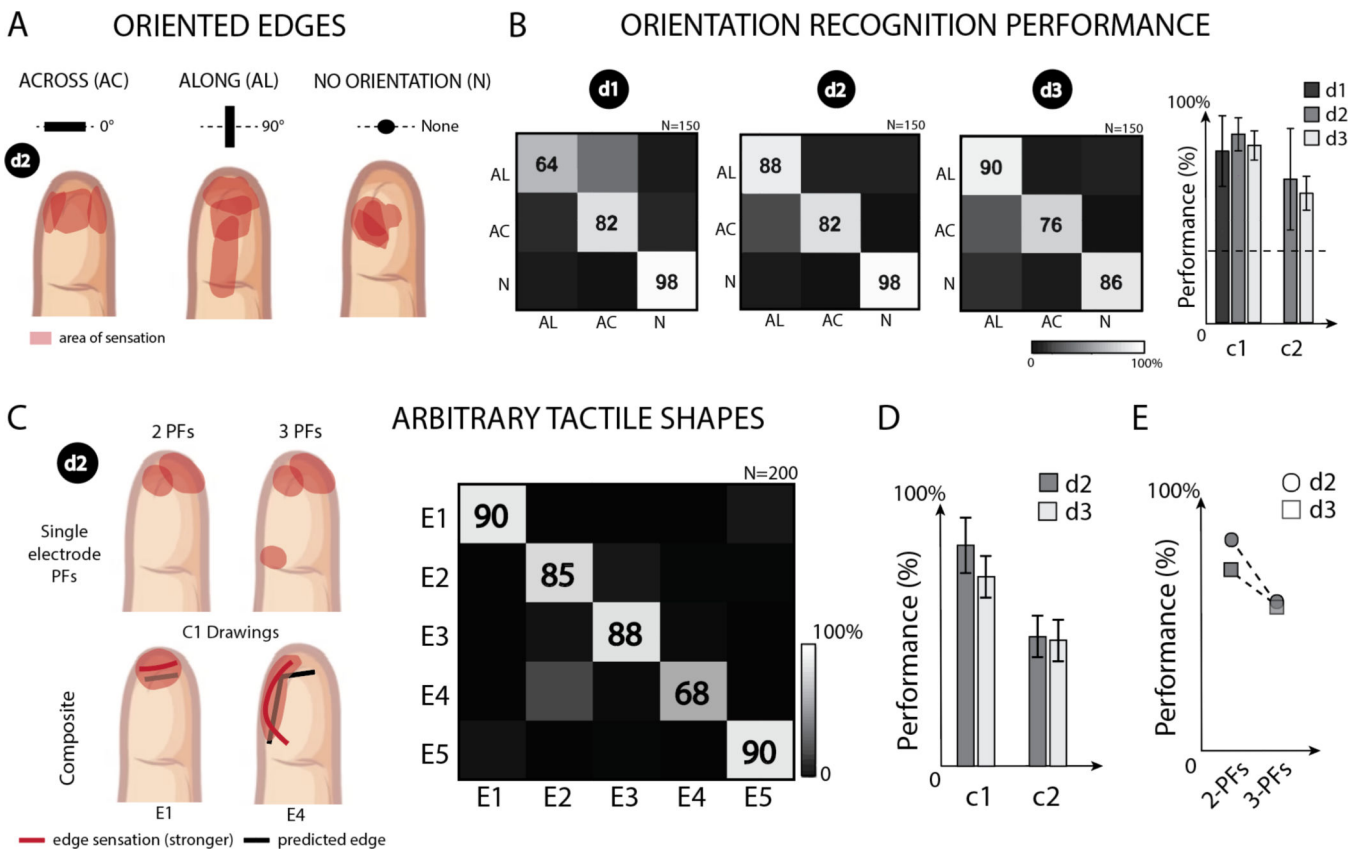


Figure 2. Multichannel ICMS of S1 evokes sensation of arbitrary tactile edges.

A| Schematic of spatial organization of the stimulated electrodes evoking sensations on digit d2 (indicated in black circle) of Along (AL), Across (AC) and No orientation (N) edges on the microelectrode array and their corresponding projected fields. **B|** Orientation identification accuracy for each digit. Columns correspond to the true label of the stimulated set of electrodes, and rows to the sensation identified by the participant. Numbers in the boxes indicate the percent of trials for which this stimulus was identified correctly. n=150. Data from c1. On the right, overall orientation recognition performance for each digit in both participants. Error bars indicate confidence intervals. **C|** (right) Examples of two-electrode combinations PFs used to create arbitrary tactile shapes on d2. Shaded area represents the area of sensation. Thick lines indicate the zone of strong sensation. Black lines are the predictions of the evoked sensation from the single electrode PFs, assuming linear summation. Composite percepts reported by c1 when stimulated with the specific combination. (left) Identification performance for five arbitrary tactile shapes, combining two PFs on d2, randomly presented to c1. Numbers in the boxes indicate the percent of successful identifications per condition. **D|** Arbitrary shape recognition broken down by participant (c1 and c2) and digit (d2, d3) (bottom left). Chance threshold: c1 – 20% and c2 – 33%. **E|** Arbitrary shape recognition broken down by ICMS-evoked edge complexity (two and three combined PFs). Chance threshold: 2PFs – 20% and 3PFs – 25%. (bottom right). Fisher-Exact-test. Data from c1.

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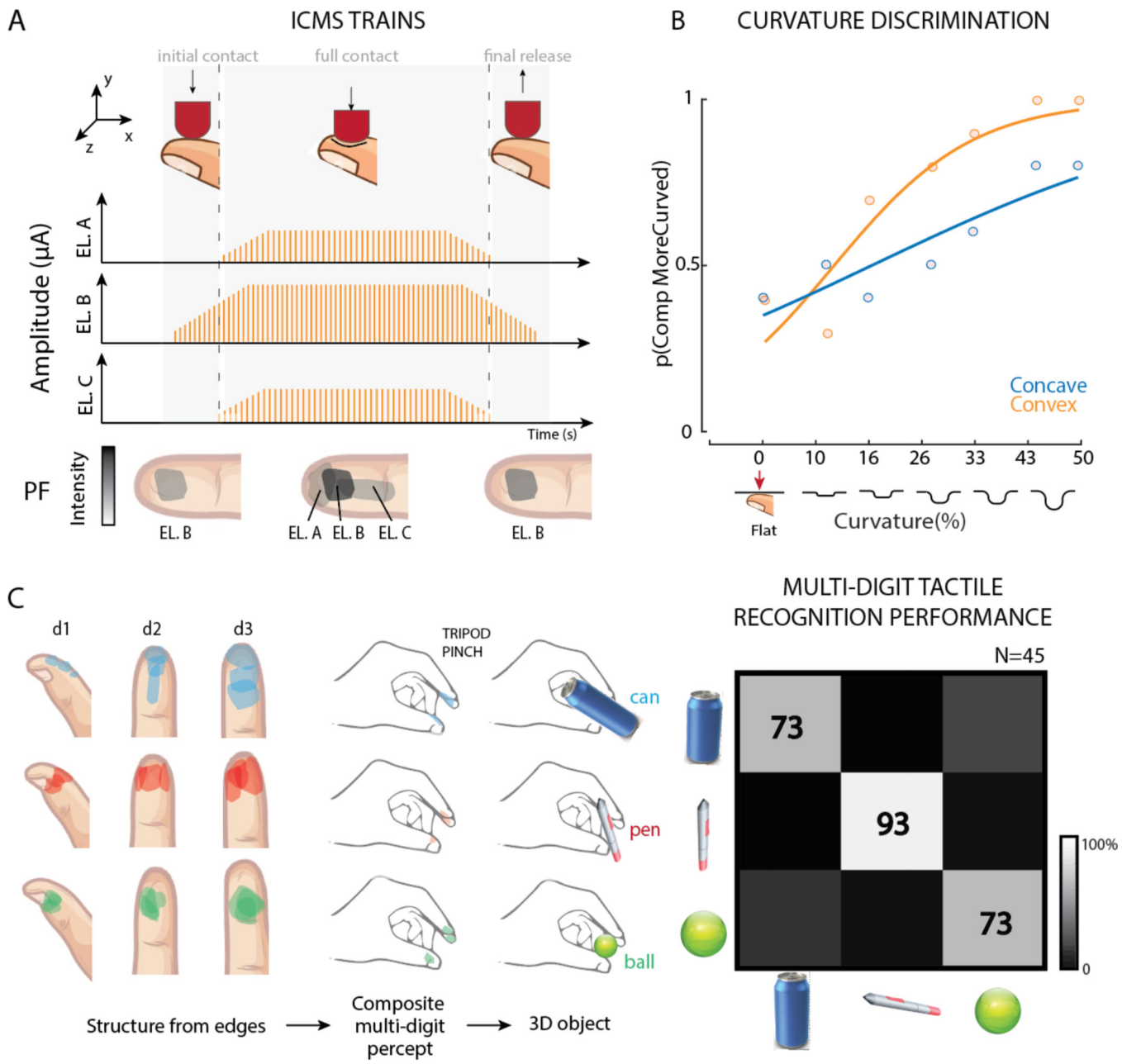


Figure 3. Tactile curvature and composite multi-digit tactile percepts via ICMS.

A| Edge curvature encoded through spatiotemporal pattering of ICMS via the modulation of the amplitude and synchrony between the activated channels. **B|** Curvature discrimination performance for concave and convex stimuli (curvature from 0% - flat to 50% - most curved). n=65. **C|** Combinations of PFs simultaneously activated on three different digits (d1, d2 and d3). PFs arise from single channel stimulation. Conditions represent stimuli on two perpendicular axes (Across and Along) and one condition with no orientation. mICMS evoked sensations on all three digits simultaneously. Types of real grasped objects perceived by c1 (Can, Pen and Ball). Multidigit identification performance. n=45. Numbers in the

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boxes indicate the percent of successful identifications per condition. Fisher-Exact-test. Data from c1.

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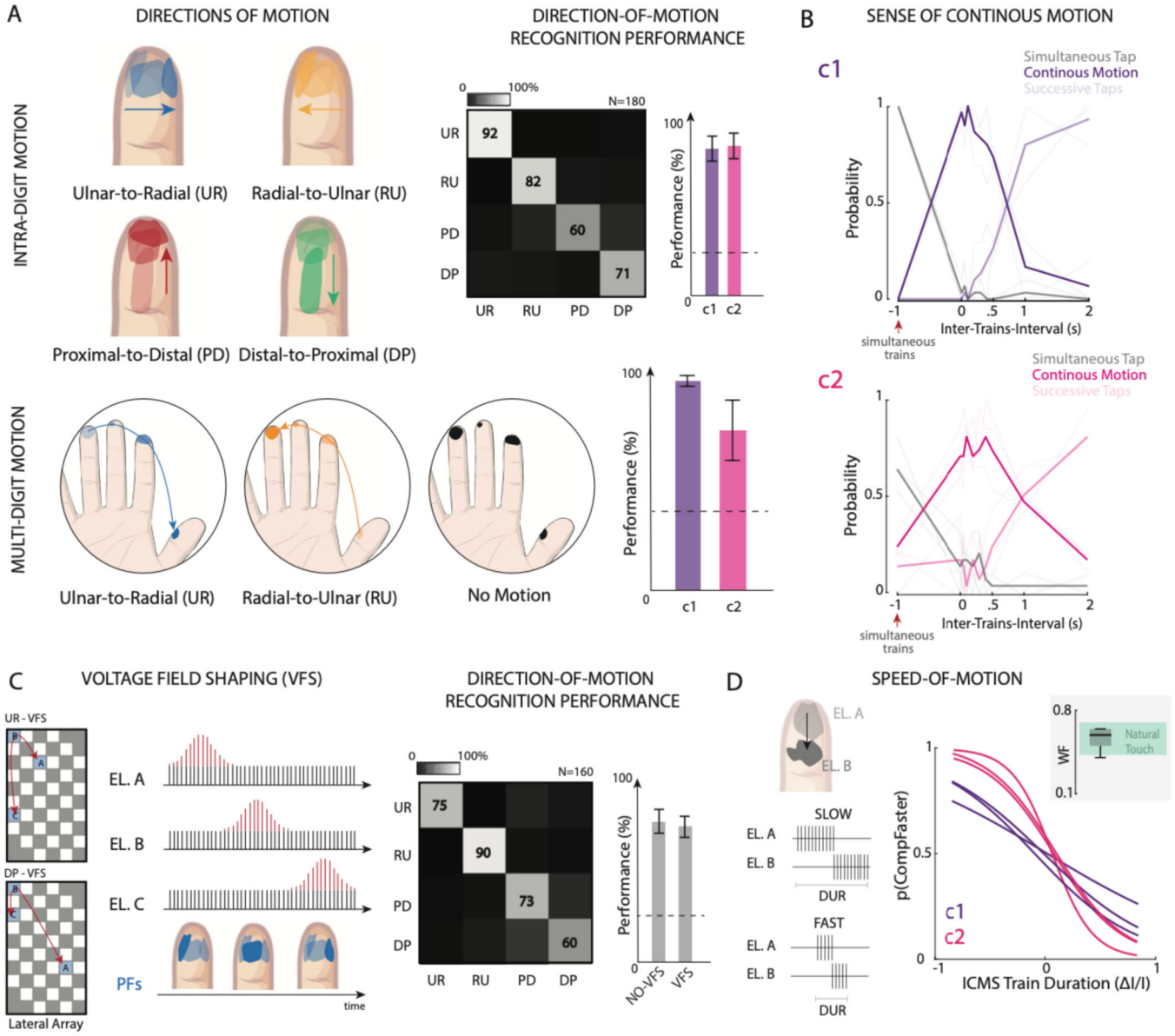


Figure 4. Patterned ICMS of S1 evokes sensations of apparent motion.

A| (top) Combinations of PFs activated sequentially in four different directions (Ulnar-to-Radial, Radial-to-Ulnar, Proximal-to-Distal and Distal-to-Proximal). Direction-of-motion recognition performance. Numbers in the boxes indicate the percent of successful identifications per condition. n=180. Performance broken down by participant. Chance threshold: 25%. (bottom) Multi-digit motion recognition among the two radial-ulnar directions (UR-RU) and no motion in c1 (n=120) and c2 (n=75). Chance threshold: 33%. **B|** Perception of continuous motion. Thick colored lines represent the probability of a sense of continuous motion; Light colored lines represent a sense of intermittent motion (taps); and Black lines represent no motion perception (simultaneous taps). n=300. c1 and c2 data. **C|** Direction-of-motion recognition performance via amplitude modulation of mICMS. Numbers in the boxes indicate the percent of successful identifications per condition. n=160. Performance broken down by stimulation condition (NO-VFS and VSF; right). Chance

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threshold: 25%. Fisher-Exact-test. Data from c1. **D|** Encoding motion speed via mICMS. Speed discrimination performance for both c1 and c2. All the Weber fractions (WFs) are displayed in relation to the natural range of skin perception(37) (green zone).

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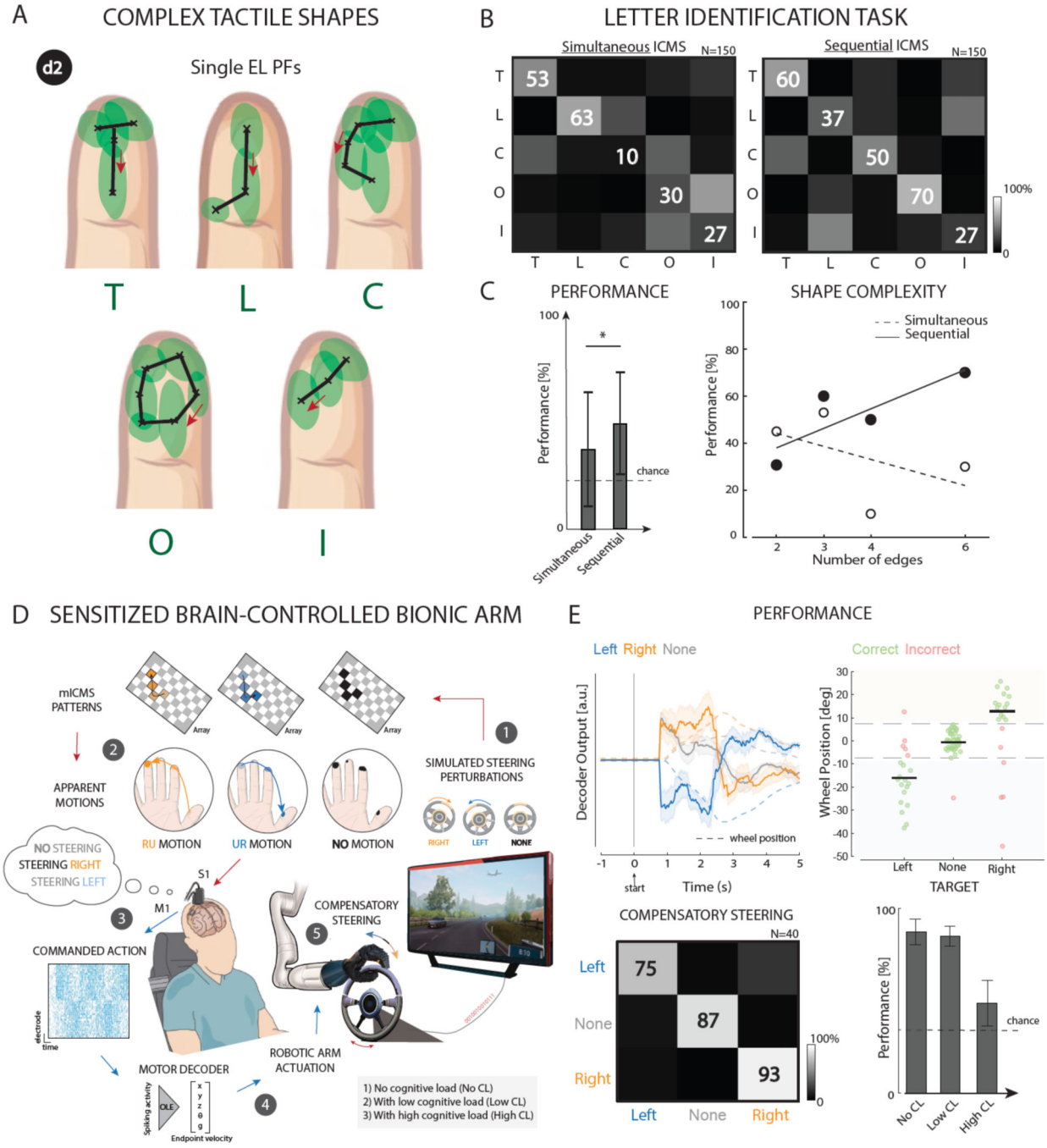


Figure 5. Sequential versus simultaneous mICMS for encoding complex tactile shapes and closed-loop functional control of a sensitized brain-controlled bionic arm.

A| Selection of PFs creating letters on d2. x indicates single electrode PF centroid. Red arrows show the order of sequential electrode activation. **B|** Letter recognition task via sequential or simultaneous ICMS. Numbers in the boxes indicate the percent of successful identifications per condition. n=150. Data from C1. **C|** Identification performance according to shape complexity (right) and direct comparison between sequential and simultaneous ICMS (left). Fisher-Exact-test *p<0.05. **D|** Schematic of the closed-loop functional task.

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CL=cognitive load. **E|** Brain decoder outputs while performing the task (No CL) are depicted (top, left). Colors represent compensatory steering directions. Correct/Incorrect trials are reported for each condition (top, right). Numbers in the boxes indicate the percent of successful steering direction per condition (bottom, left). Performance broken down by CL (bottom, right). Chance threshold: 33%. n=40. $\chi 2$ test Error bars indicate confidence intervals. Data from c1.